Institut Universitaire des Sciences

Faculté des Sciences et Technologies (FST)

TD_4 – Systèmes 1

Rapport du travail de Laboratoire #4

• **Présenté par :** Kenley CHERY

• **Niveau :** 3ème année de Licence

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Date : 10 Novembre 2026

L’objectif de ce TD est de :

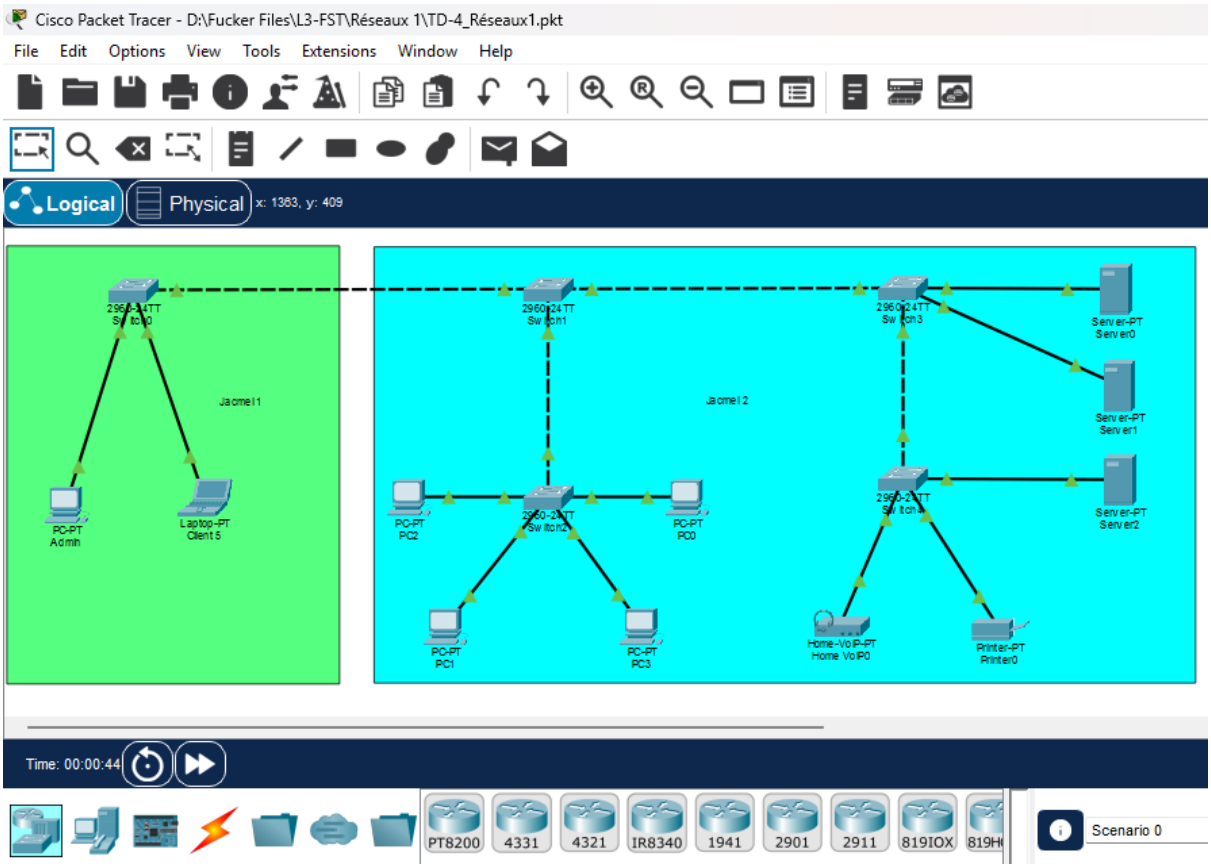
1- Découvrir la configuration de base d’un switch et d'un routeur.
2- Attribuer des adresses IPv4 et IPv6 aux périphériques.
3- Tester la connectivité entre les hôtes.
4- Utiliser le Mode Simulation pour analyser le trafic réseau

Démarche suivie :

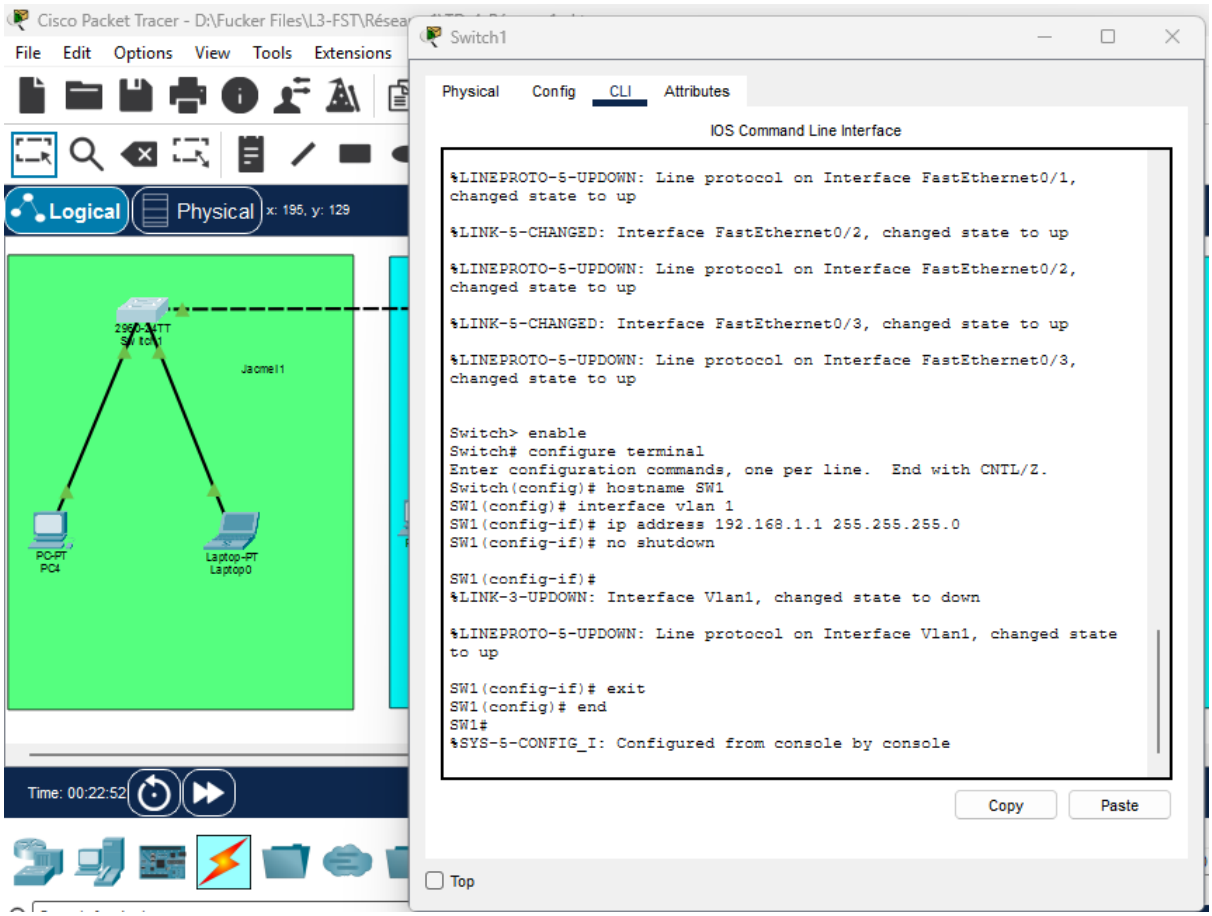
1- Création de la topologie
2- Configuration des switchs en IPv4
3- Configuration des PC en IPv4
4- Tests de connectivité
5- Intégration et configuration du Routeur
6- Validation finale de la connectivité inter-réseau

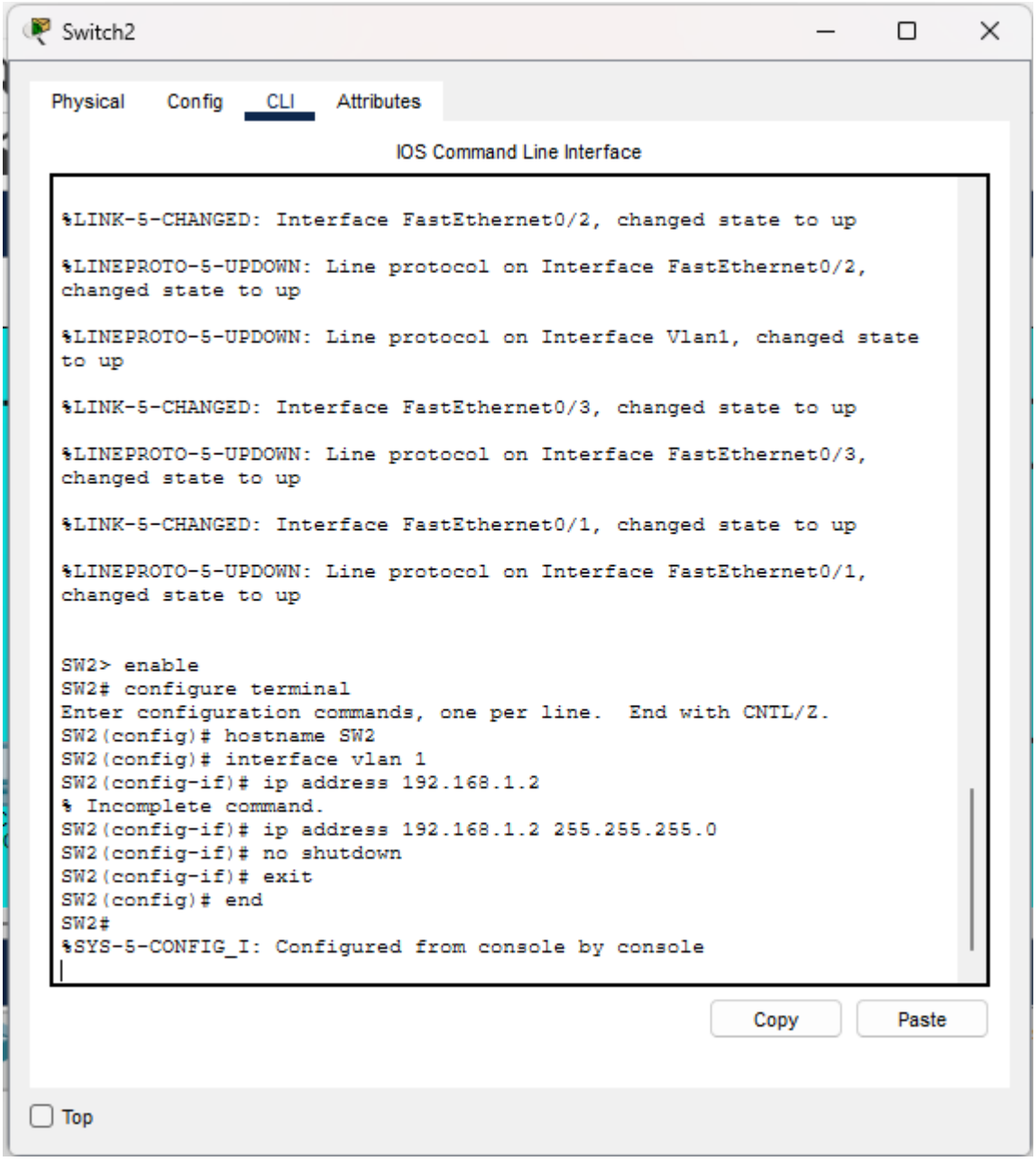
Étape 1 –

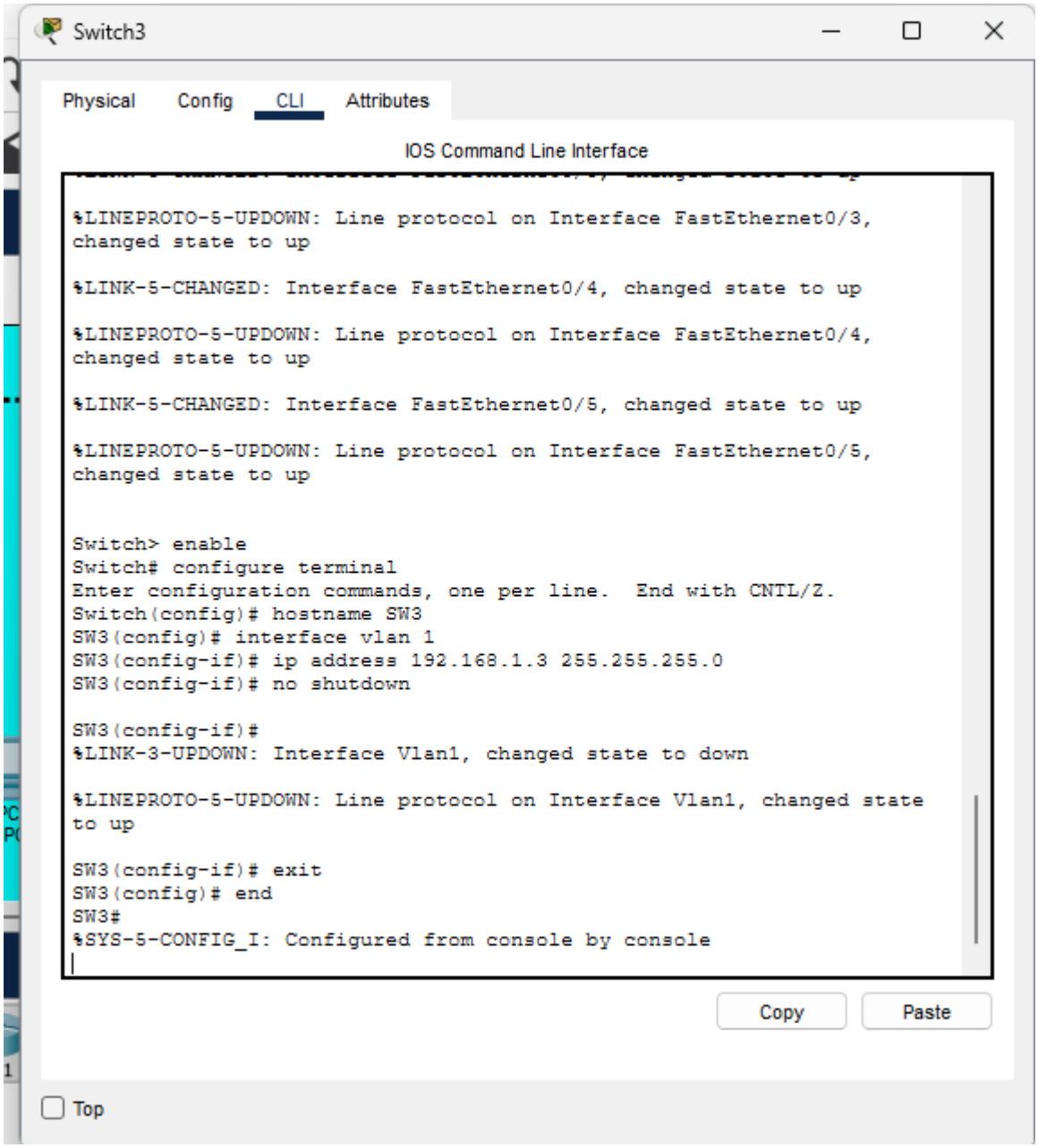
1- Création de la topologie

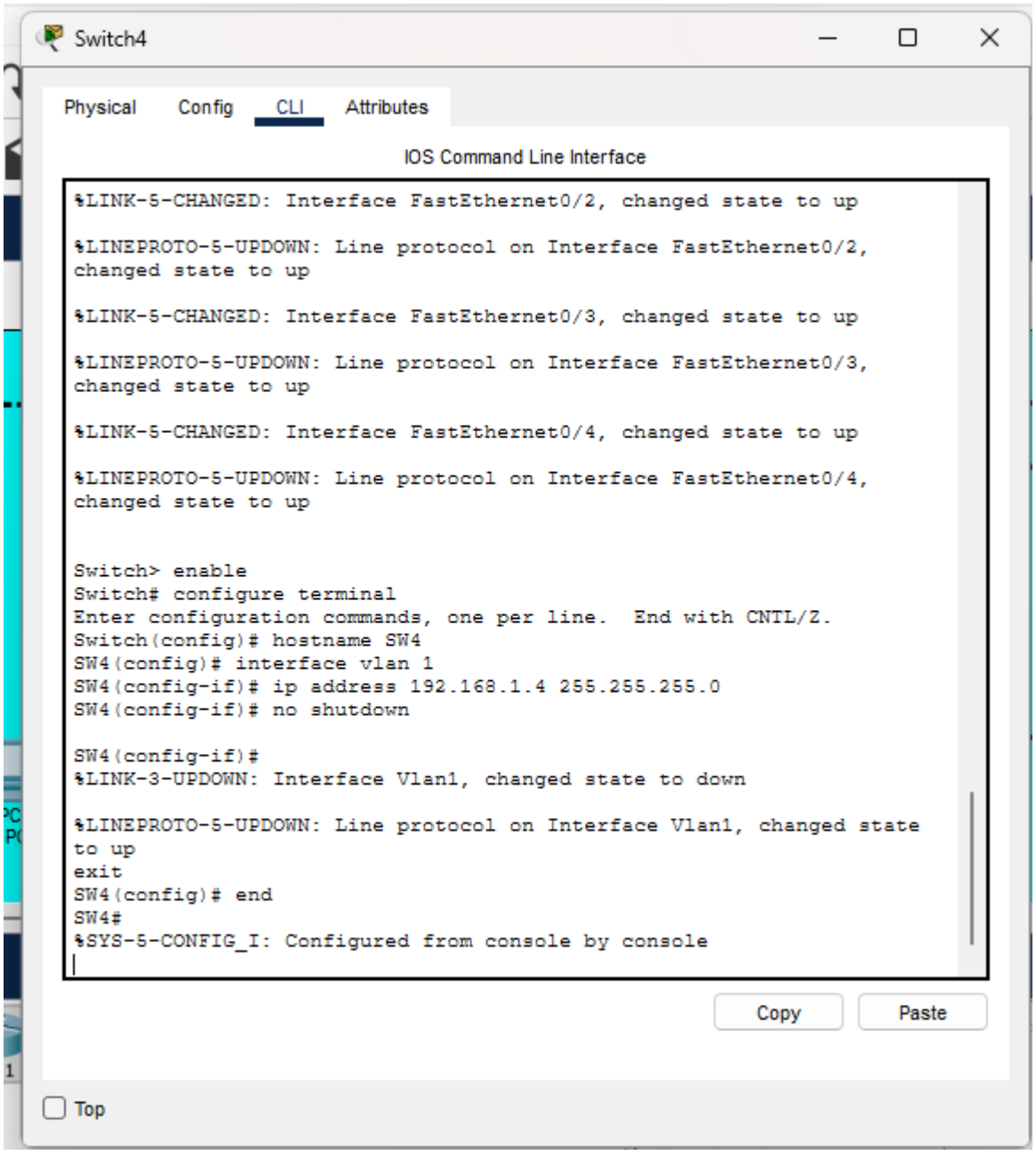


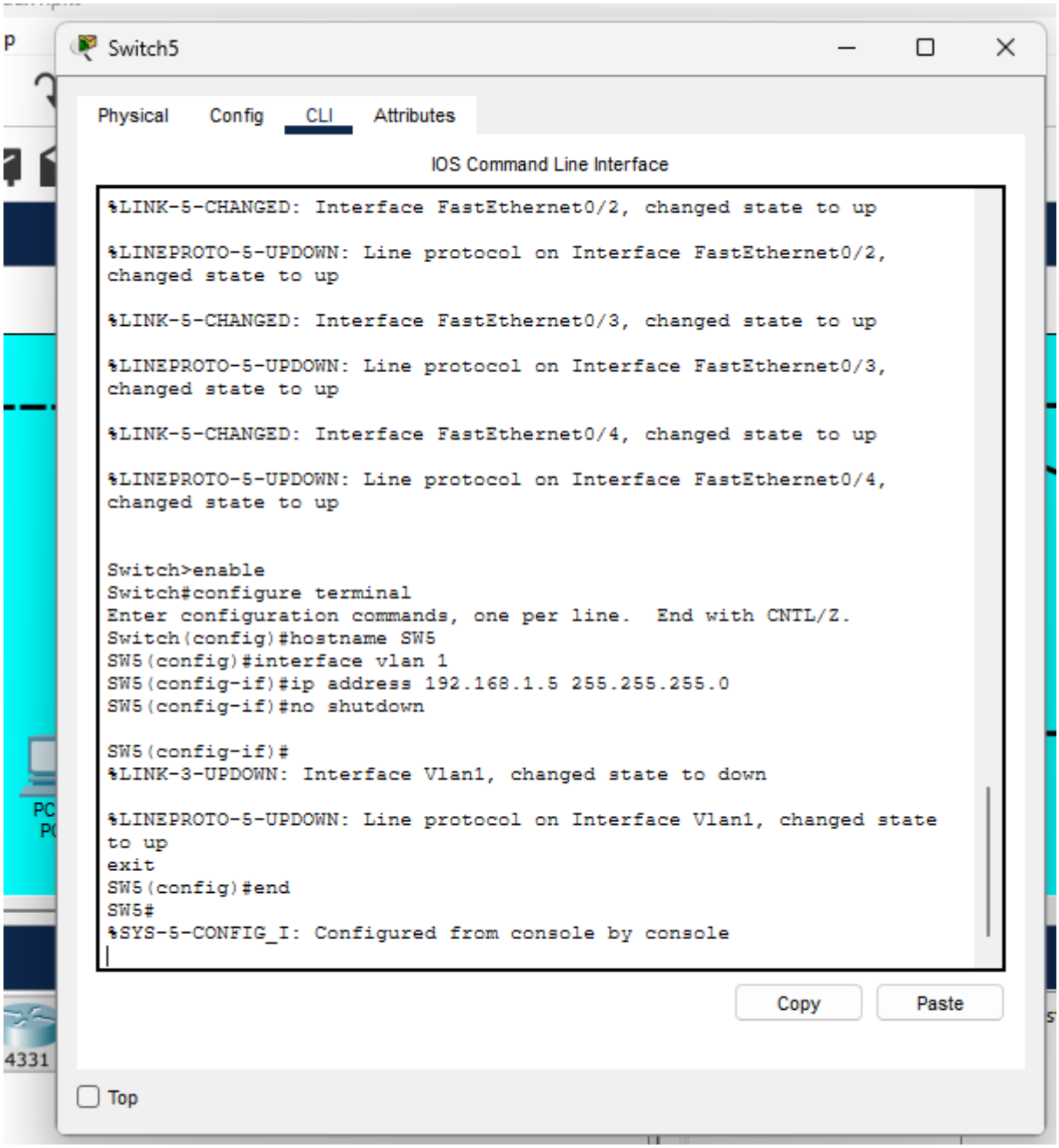
2- Configuration des switchs IPv4


Cisco Packet Tracer - D:\Fucker Files\L3-FST\Réseau
File Edit Options View Tools Extensions
Logical
Physical
x: 195, y: 129
Jacmel1
PC-PT PC4
Laptop-PT Laptop0
Time: 00:22:52
Switch1
Physical Config CLI Attributes
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
Switch> enable
Switch# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)# hostname SW1
SW1(config)# interface vlan 1
SW1(config-if)# ip address 192.168.1.1 255.255.255.0
SW1(config-if)# no shutdown
SW1(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
SW1(config-if)# exit
SW1(config)# end
SW1#
%SYS-5-CONFIG_I: Configured from console by console
Copy
Paste
Top

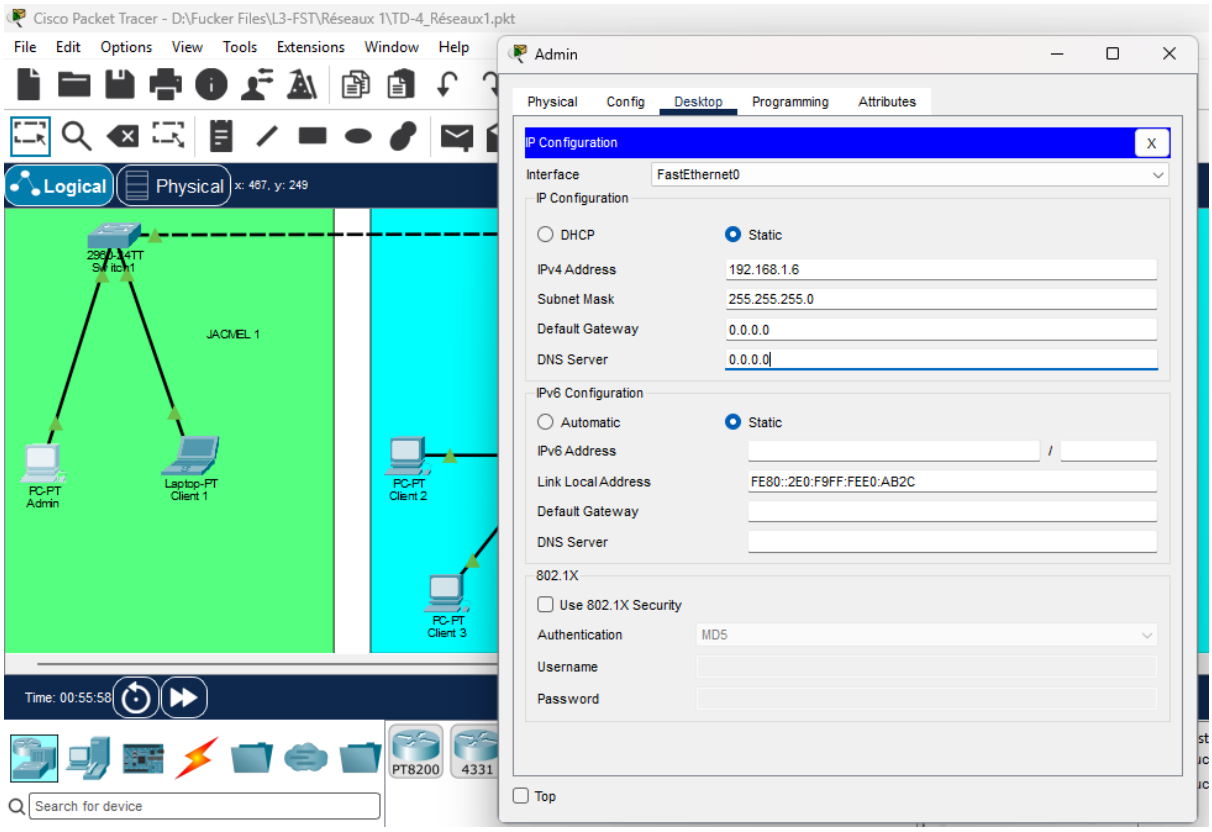

Switch2
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
SW2> enable
SW2# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SW2(config)# hostname SW2
SW2(config)# interface vlan 1
SW2(config-if)# ip address 192.168.1.2
% Incomplete command.
SW2(config-if)# ip address 192.168.1.2 255.255.255.0
SW2(config-if)# no shutdown
SW2(config-if)# exit
SW2(config)# end
SW2#
%SYS-5-CONFIG_I: Configured from console by console
Copy
Paste
Top

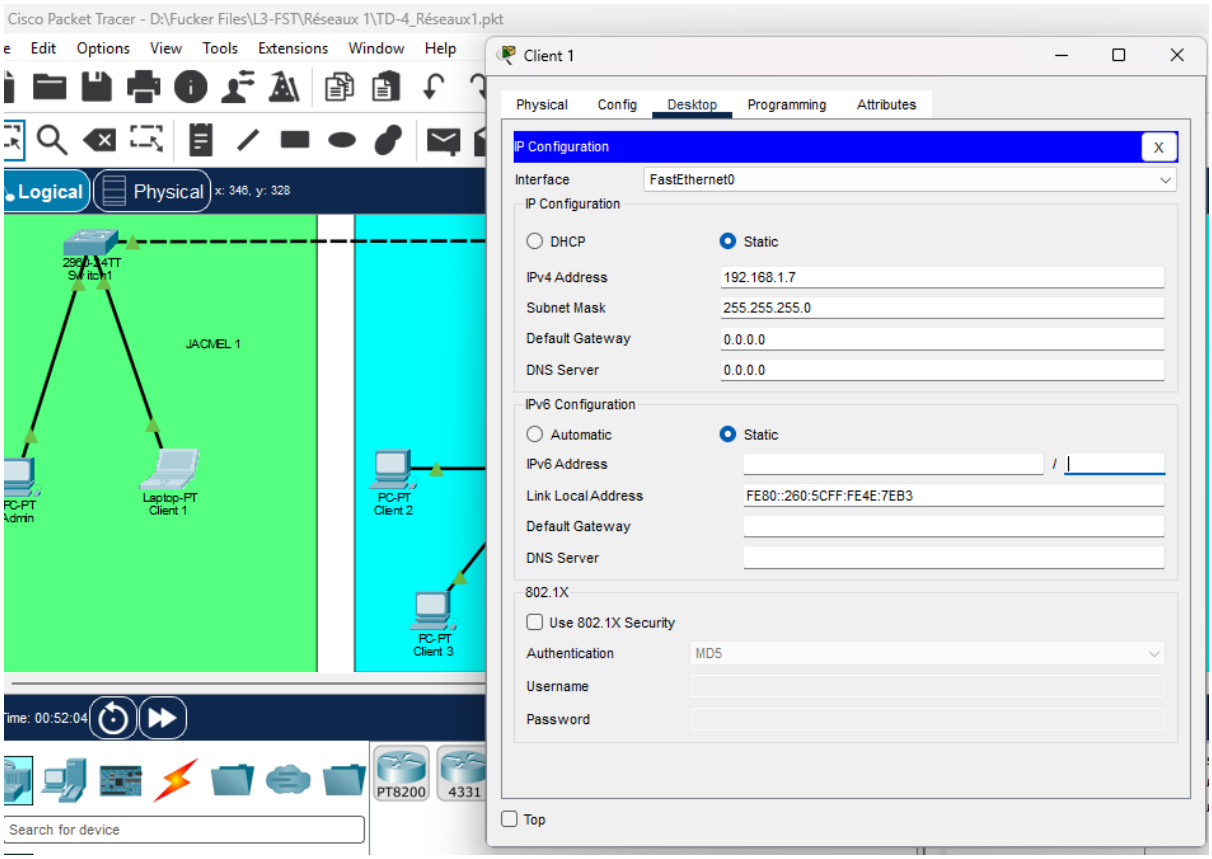

Switch3
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
Switch> enable
Switch# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)# hostname SW3
SW3(config)# interface vlan 1
SW3(config-if)# ip address 192.168.1.3 255.255.255.0
SW3(config-if)# no shutdown
SW3(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
SW3(config-if)# exit
SW3(config)# end
SW3#
%SYS-5-CONFIG_I: Configured from console by console
Copy
Paste
Top

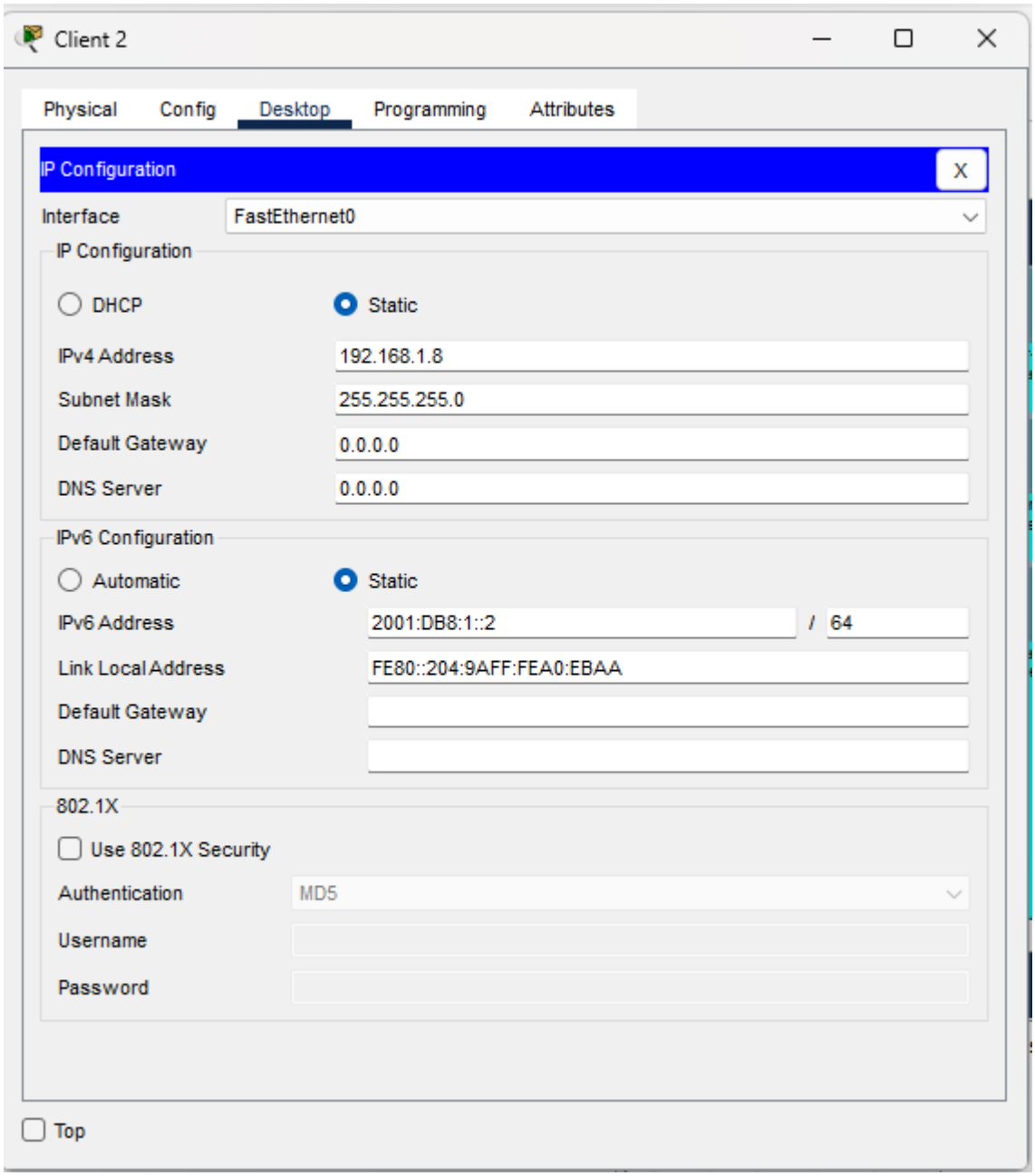

Switch4
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
Switch> enable
Switch# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)# hostname SW4
SW4(config)# interface vlan 1
SW4(config-if)# ip address 192.168.1.4 255.255.255.0
SW4(config-if)# no shutdown
SW4(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
exit
SW4(config)# end
SW4#
%SYS-5-CONFIG_I: Configured from console by console
Copy
Paste
Top

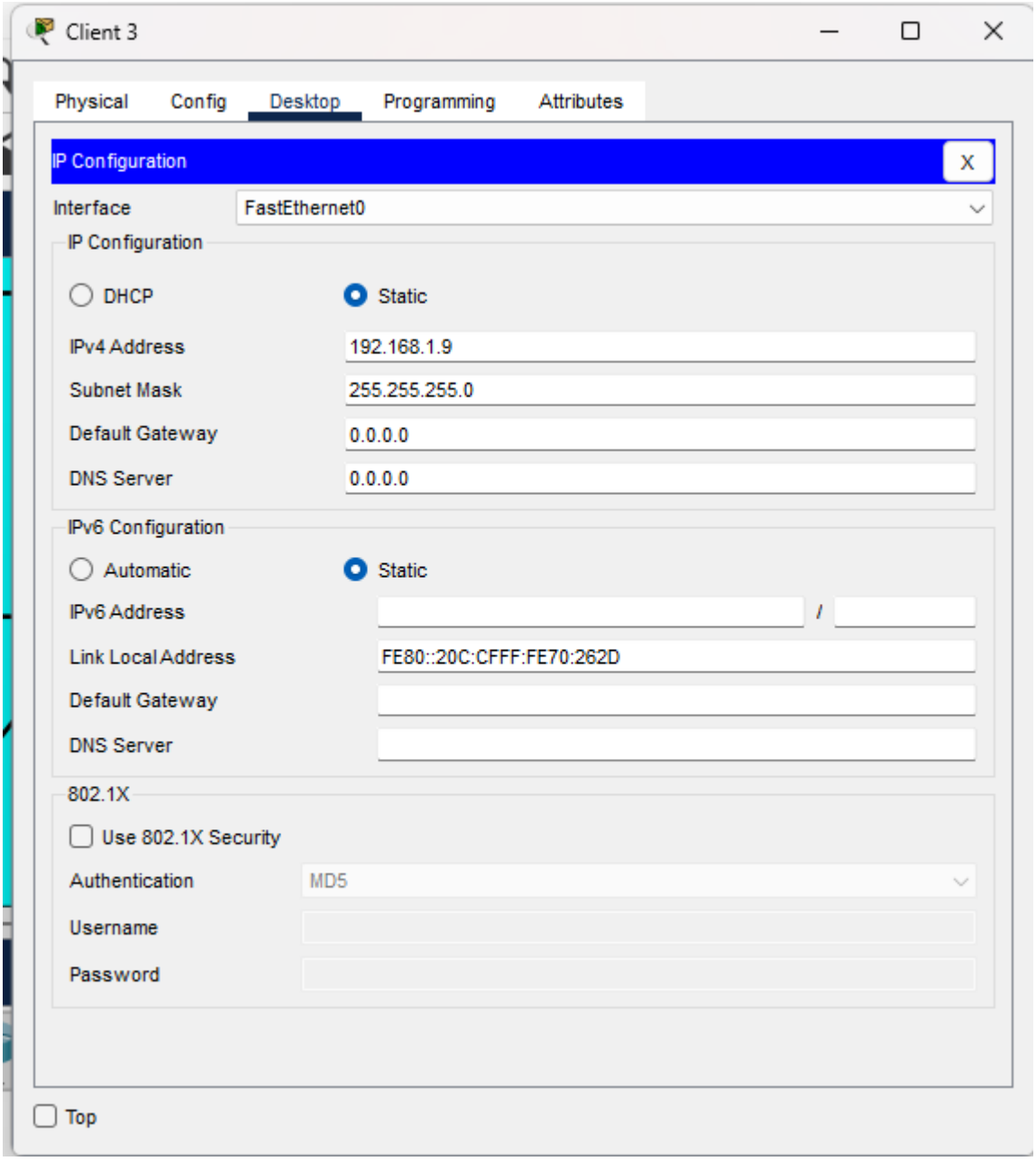


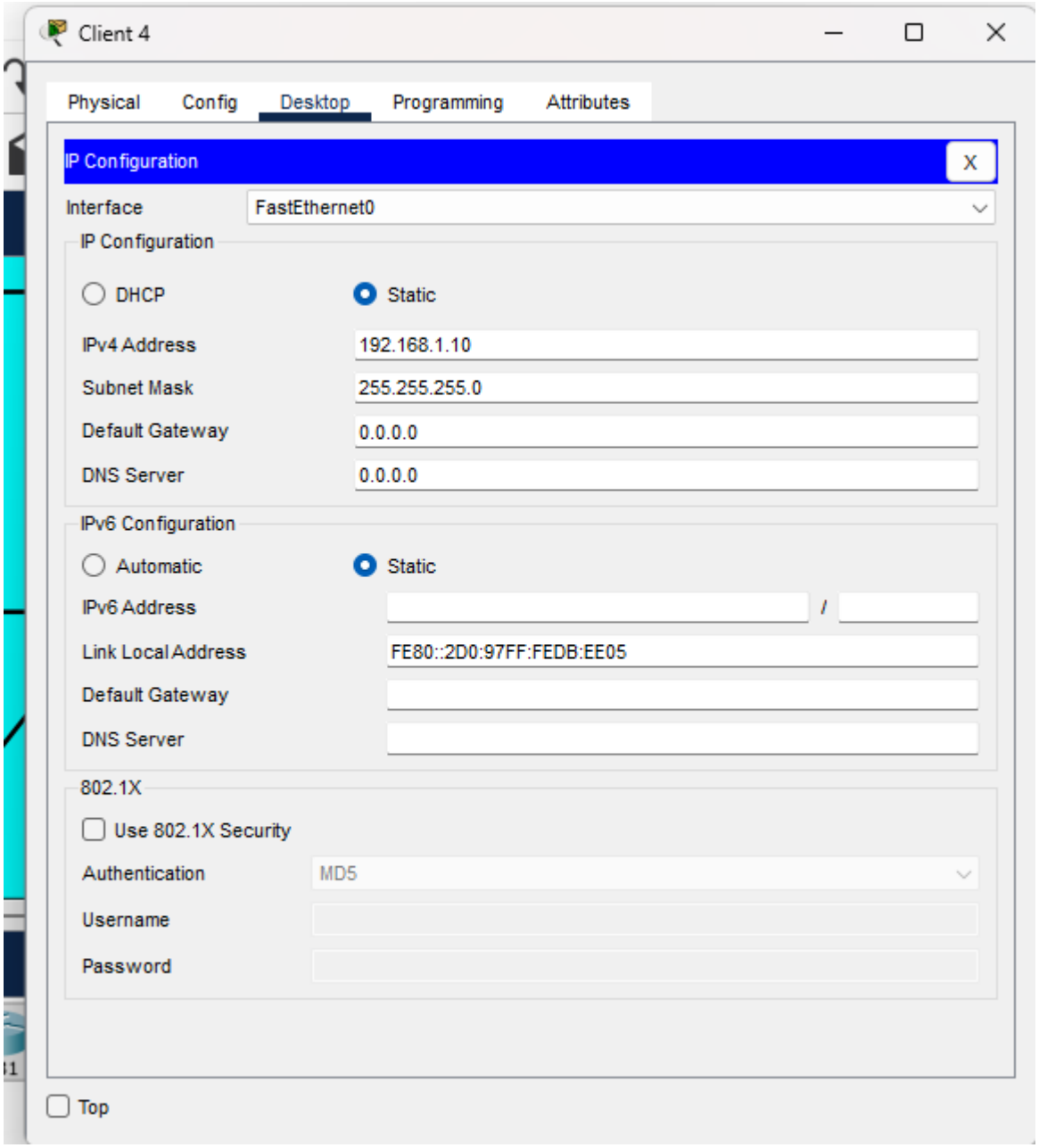
3- Configuration des PC en IPv4

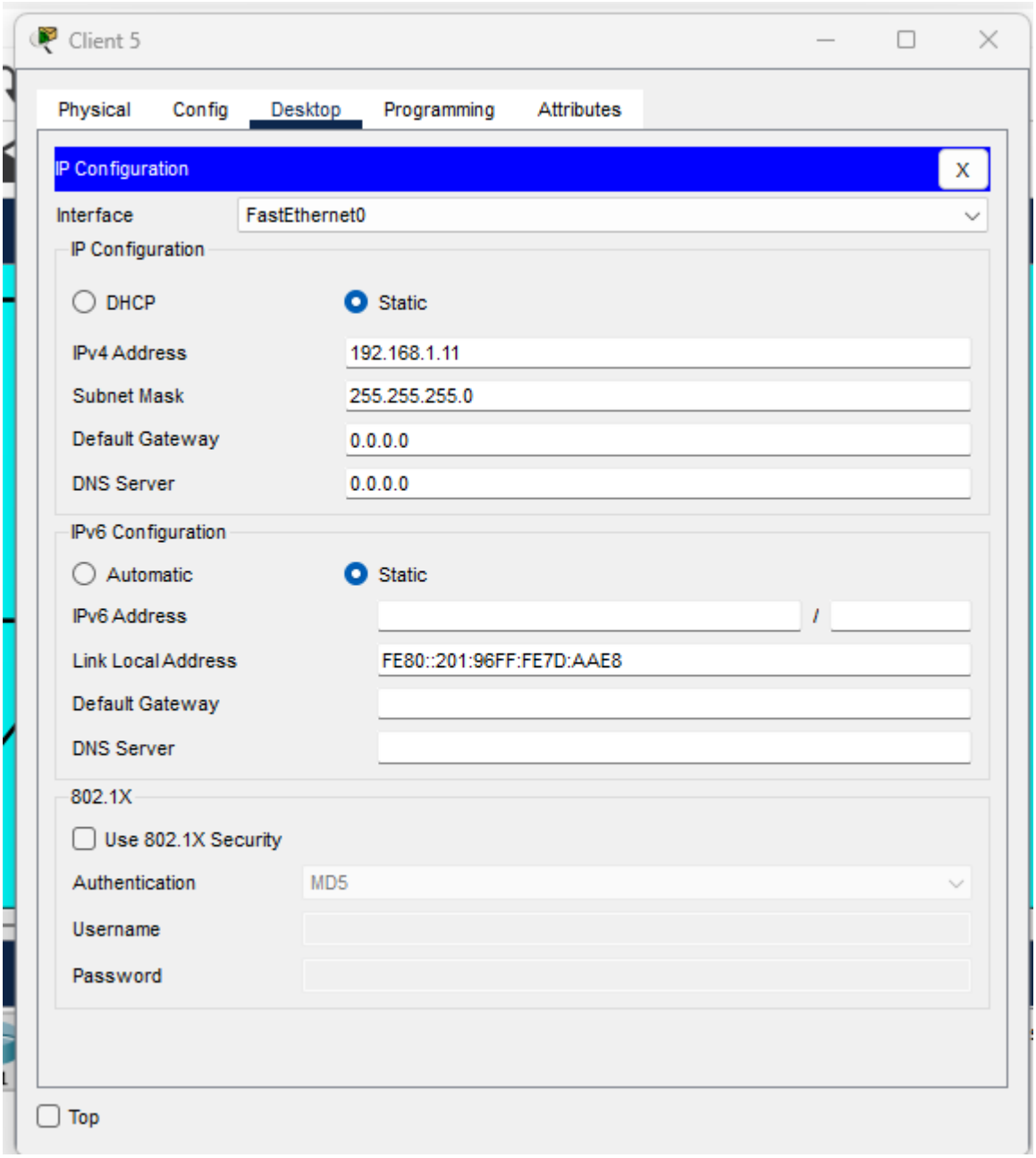

Cisco Packet Tracer - D:\Fucker Files\L3-FST\Réseaux 1\TD-4_Réseaux1.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 467, y: 249
JACMEL 1
PC-PT Admin
Laptop-PT Client 1
PC-PT Client 2
PC-PT Client 3
Time: 00:55:58
Search for device
PT8200
4331
Admin
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.6
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::2E0:F9FF:FEE0:AB2C
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


Cisco Packet Tracer - D:\Fucker Files\L3-FST\Réseaux 1\TD-4_Réseaux1.pkt
Edit
Options
View
Tools
Extensions
Window
Help
Logical
Physical
x: 346, y: 328
JACMEL 1
Laptop-PT
Client 1
PC-PT
Client 2
PC-PT
Client 3
Time: 00:52:04
PT8200
4331
Search for device
Client 1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.7
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::260:5CFF:FE4E:7EB3
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

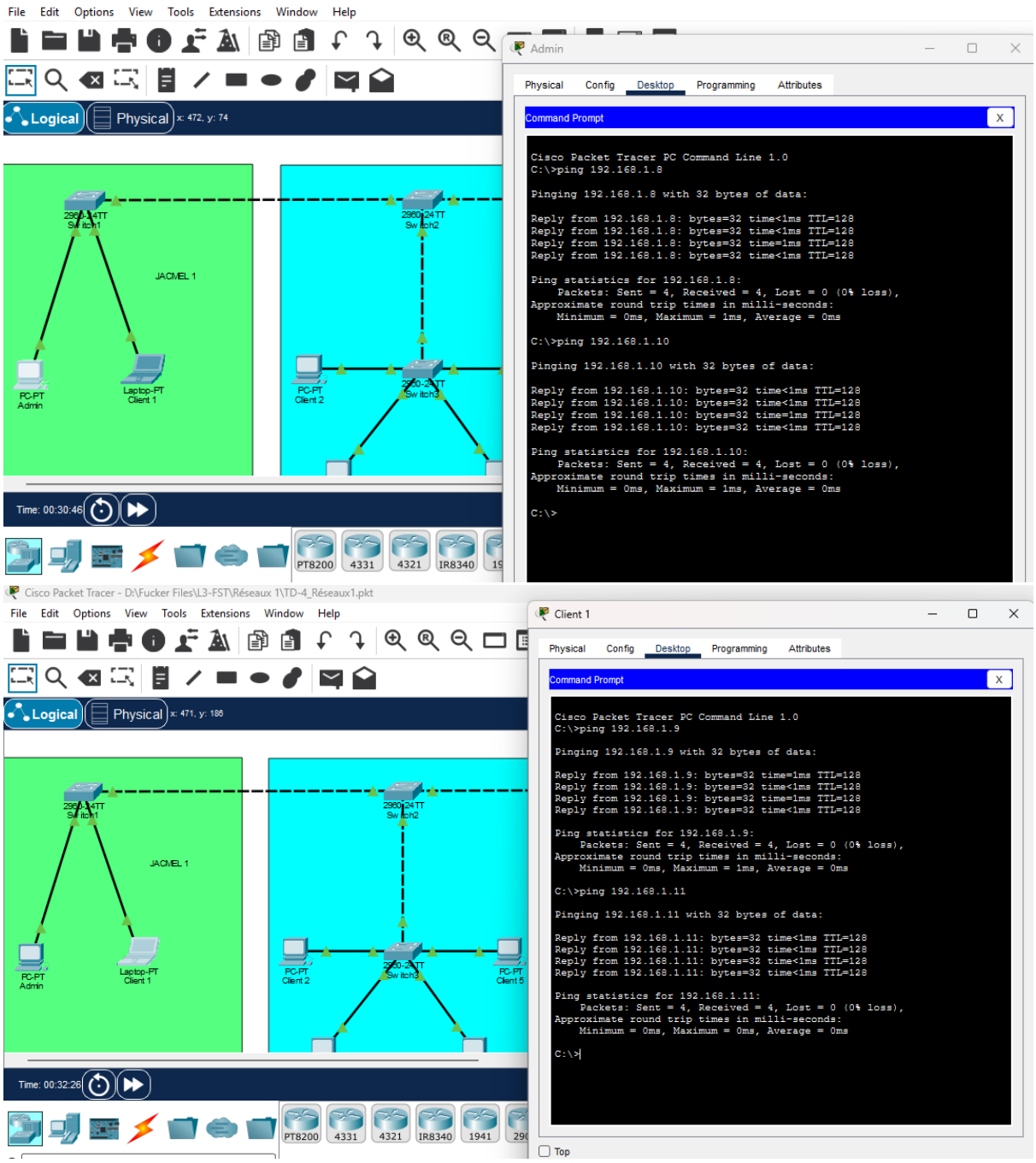

Client 2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.8
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:DB8:1::2
/
64
Link Local Address
FE80::204:9AFF:FEA0:EBAA
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


Client 3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.9
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::20C:CFFF:FE70:262D
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

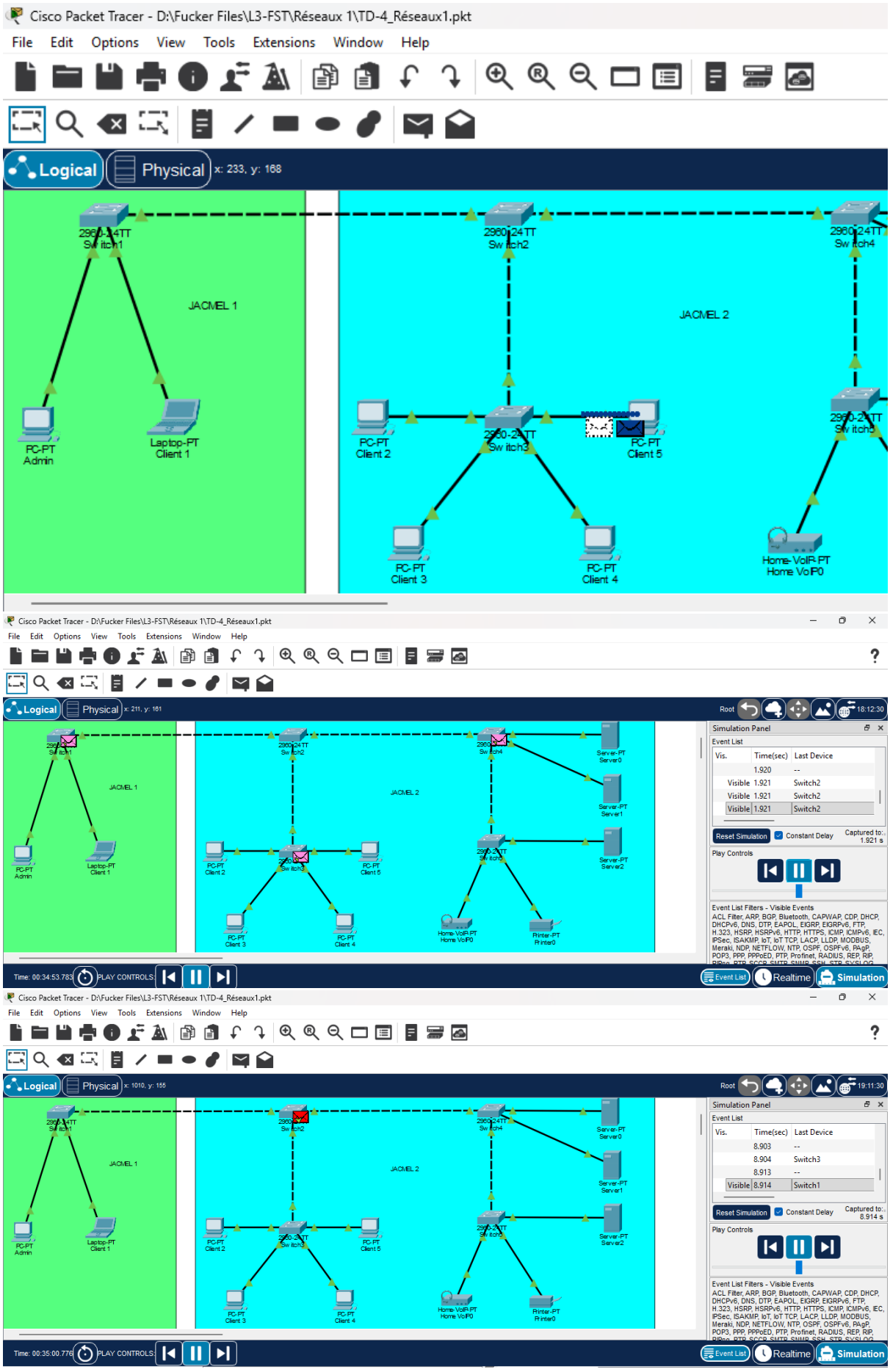

Client 4
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.10
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::2D0:97FF:FEDB:EE05
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

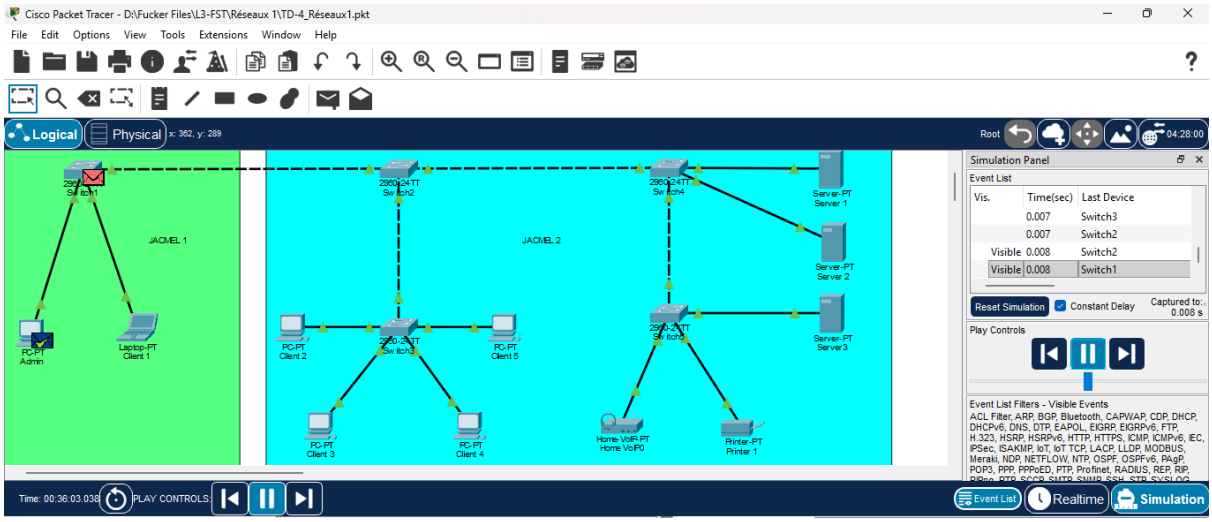


4- Test de connectivité du pc Admin et du Client 1 de Jacmel 1 avec certains pc de Jacmel 2 avec la commande ping :



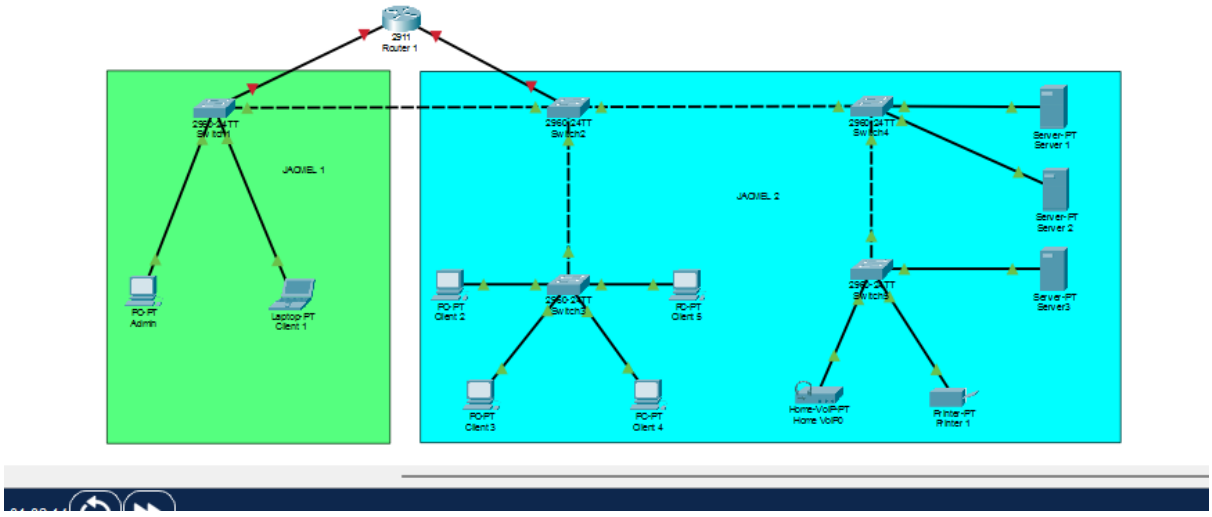
5- Passons le test au mode simulation :


Cisco Packet Tracer - D:\Fucker Files\L3-FST\Réseaux 1\TD-4_Réseaux1.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 233, y: 168
JACMEL 1
JACMEL 2
2960-24TT Switch1
2960-24TT Switch2
2960-24TT Switch4
2960-24TT Switch3
PC-PT Admin
Laptop-PT Client 1
PC-PT Client 2
PC-PT Client 5
PC-PT Client 3
PC-PT Client 4
Home-VoIP-PT Home VoIP0
Server-PT Server0
Server-PT Server1
Server-PT Server2
Printer-PT Printer0
Simulation Panel
Event List
Vis. Time(sec) Last Device
1.920 --
Visible 1.921 Switch2
Visible 1.921 Switch2
Visible 1.921 Switch2
Reset Simulation
Constant Delay
Captured to: 1.921 s
Play Controls
Event List Filters - Visible Events
ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoED, PTP, Profinet, RADIUS, REP, RIP,
Time: 00:34:53.783
PLAY CONTROLS:
Event List
Realtime
Simulation
Cisco Packet Tracer - D:\Fucker Files\L3-FST\Réseaux 1\TD-4_Réseaux1.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1010, y: 155
Simulation Panel
Event List
Vis. Time(sec) Last Device
8.903 --
8.904 Switch3
8.913 --
Visible 8.914 Switch1
Reset Simulation
Constant Delay
Captured to: 8.914 s
Play Controls
Event List Filters - Visible Events
Time: 00:35:00.776
PLAY CONTROLS:
Event List
Realtime
Simulation

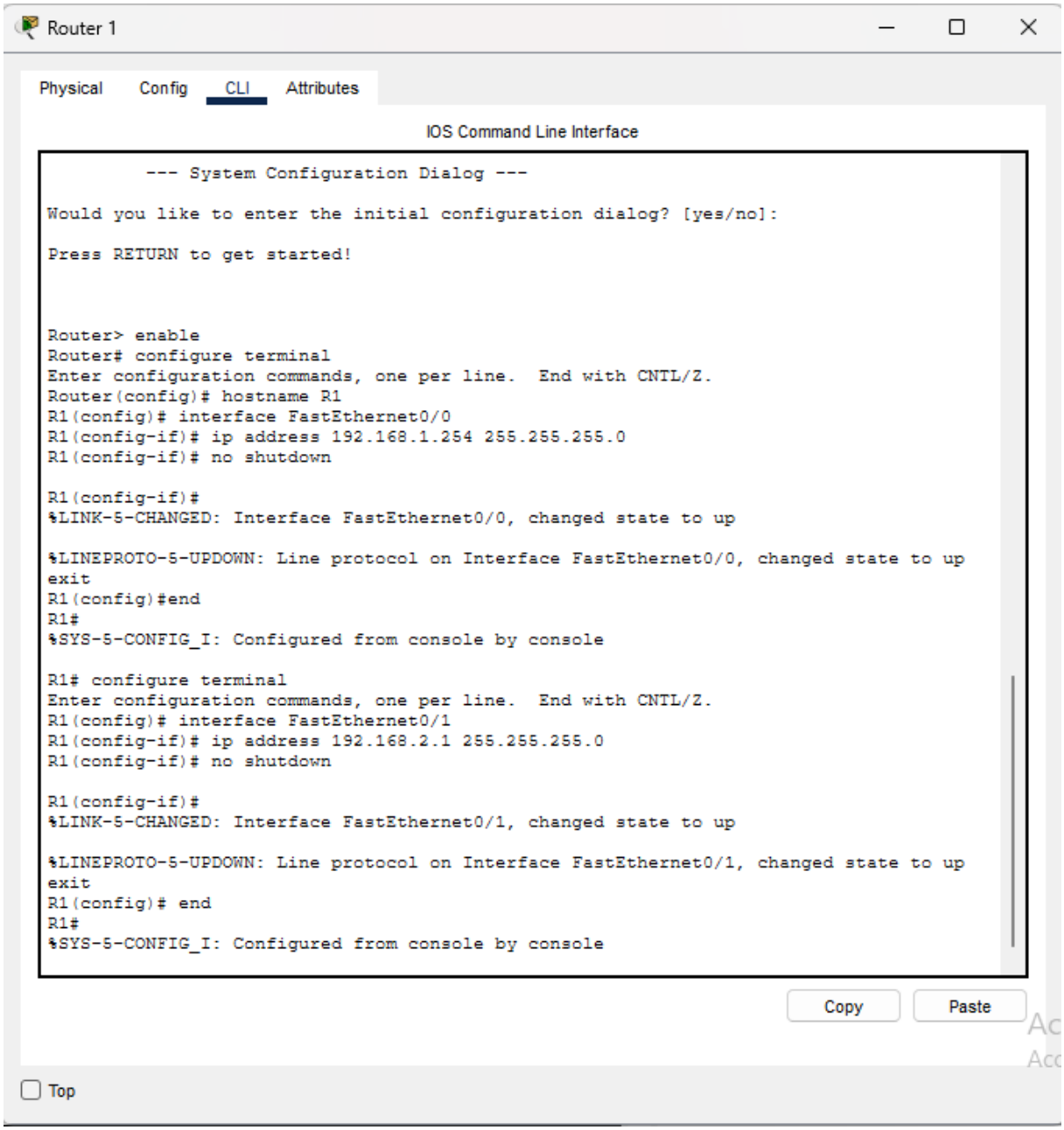


Étape 2 –

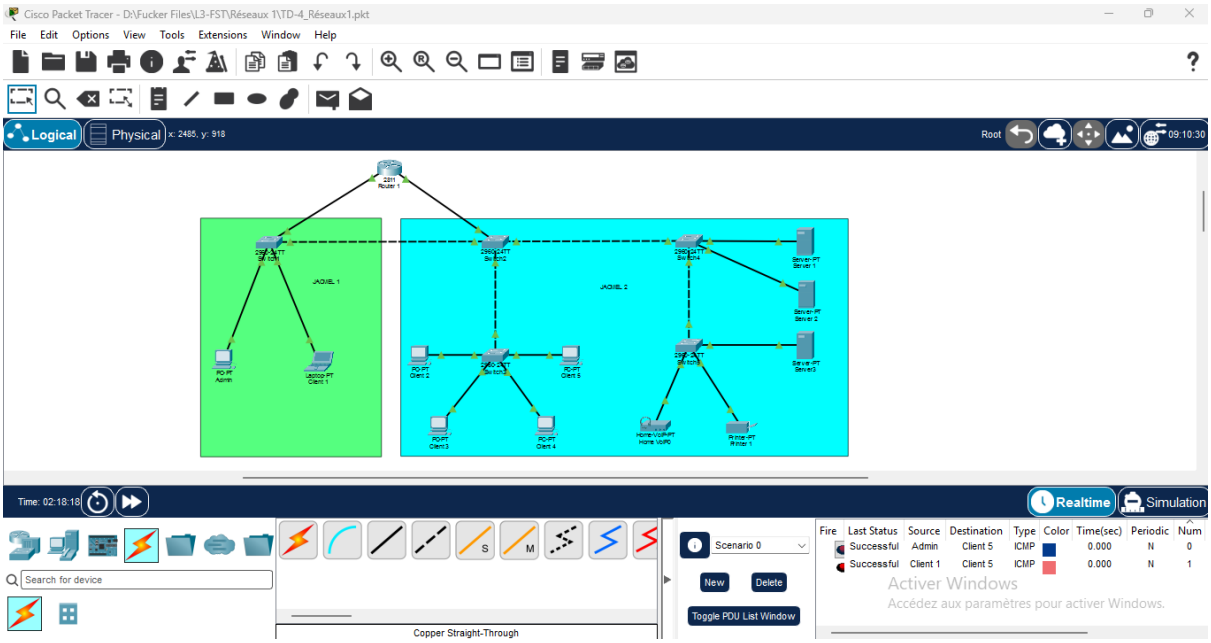
Répresentation de la Topologie



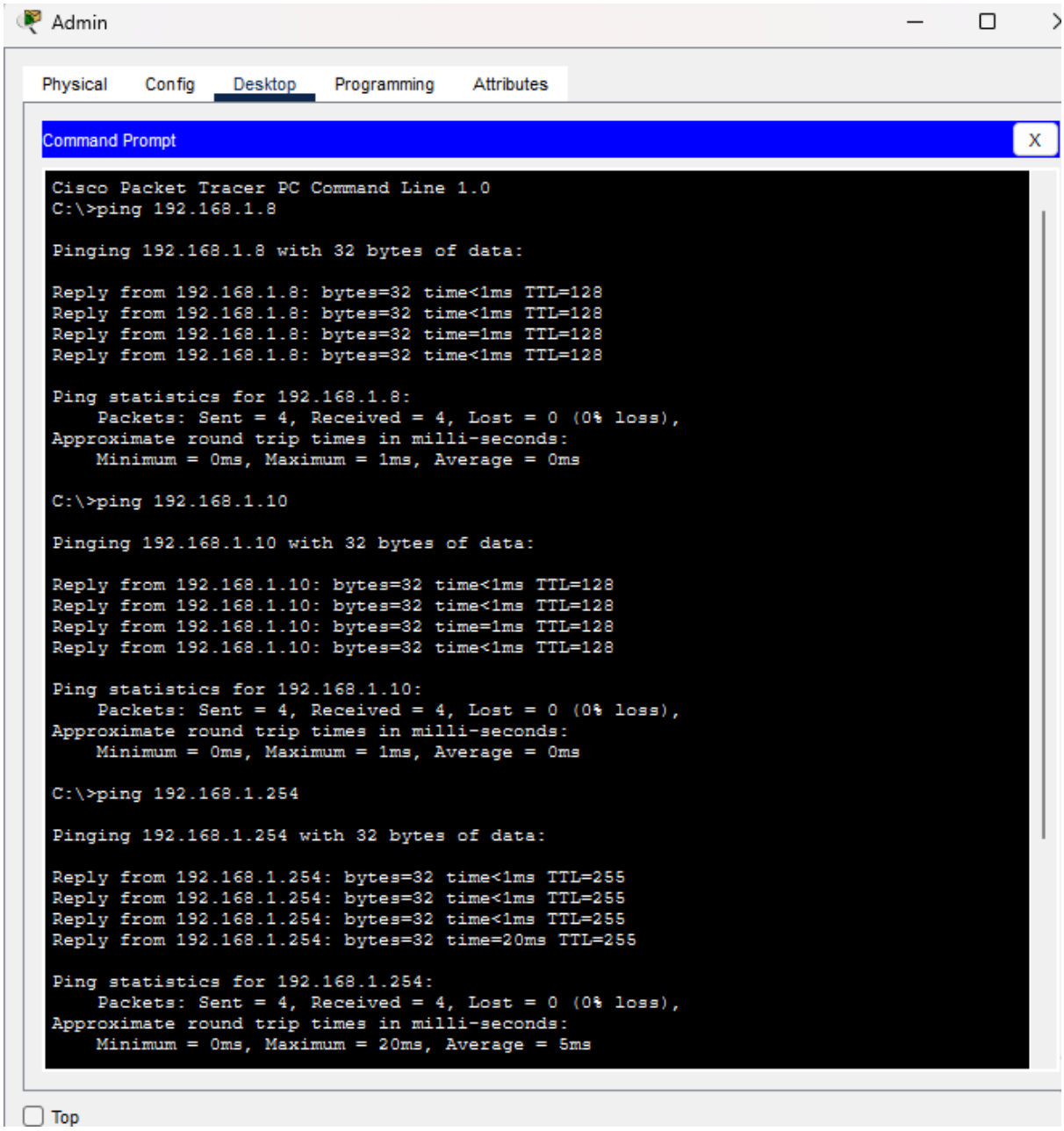
1- La configuration du Routeur



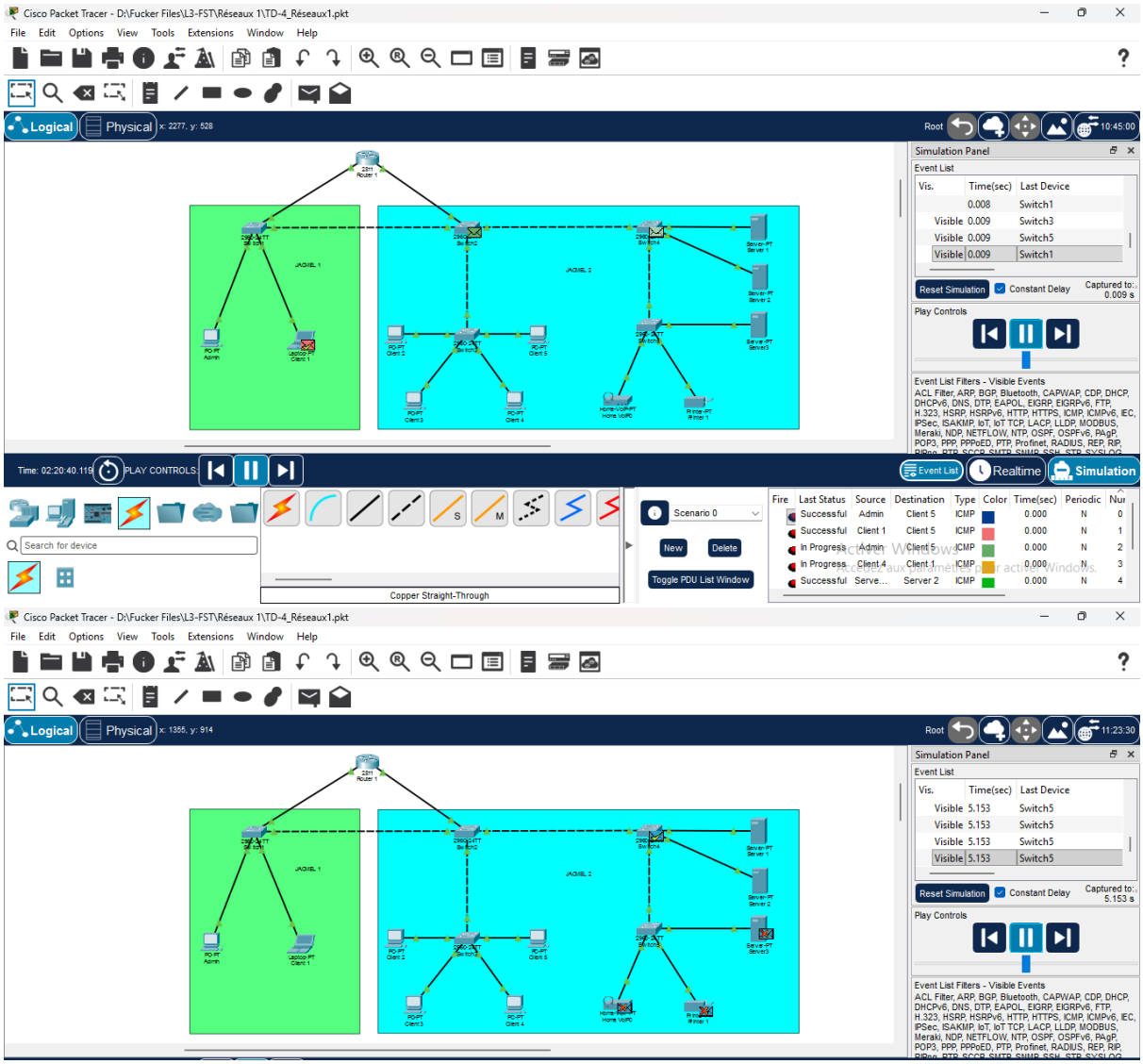
2- La Topologie après la configuration



3- Test de connectivité avec la commande ping



4- Passons le test au mode simulation :



Problèmes rencontrés et solutions apportées

- **Erreur d'attribution d'adresse globale sur un commutateur (Switch) :**
  Problème *:* Tentative initiale d'attribuer une adresse IP directement en mode de configuration globale, provoquant l'erreur % Invalid input detected at '^' marker.

  Solution : Passage par l'interface virtuelle du commutateur (interface vlan 1) pour y affecter l'adresse de gestion.

- **Erreur de syntaxe sur les interfaces du routeur :**
  Problème : Message d'erreur % Invalid interface type and number lors de l'utilisation du type GigabitEthernet.

  Solution : Identification du modèle matériel du routeur (Cisco 2811) et basculement vers la désignation d'interface correcte : FastEthernet0/0 et FastEthernet0/1.

- **Conflit de chevauchement de sous-réseau (Overlap) :**
  Problème : Message d'erreur % 192.168.1.0 overlaps with FastEthernet0/0 lors de la tentative d'attribution d'une IP de la plage 1 sur l'interface droite.

  Solution : Application d'un plan d'adressage distinct pour isoler logiquement la zone JACMEL 2 sur un sous-réseau séparé (192.168.2.0/24), respectant le rôle de routage de couche 3.

Conclusion

Ce travail de laboratoire a permis de mettre en pratique les concepts fondamentaux de l'interconnexion réseau sous Cisco Packet Tracer. La configuration de base des commutateurs et du routeur a mis en évidence la frontière claire entre les équipements de couche 2. L'utilisation concrète de la ligne de commande (CLI) a renforcé notre autonomie face aux messages d'erreur de syntaxe et aux contraintes logiques d'adressage (comme le chevauchement de sous-réseau). Enfin, l'analyse dynamique du trafic en mode Simulation a validé la bonne transmission des données et le rôle de passerelle joué par le routeur pour interconnecter les réseaux JACMEL 1 et JACMEL 2.